

Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

(6.1a)

$$\mathbf{val}^\circ = (\mathbf{embed}^\circ \mathbf{wrap}) \cup (\mathbf{op}^\circ \neg \mathbf{Digit}(\mathbf{val}^\circ) \mathbf{snoc})$$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

the converse of $\mathbf{val}$, which reads digits as a number

(a)
(b)
(c)
(d)
(e)

(a)	$\mathbf{val}^\circ$	$\mathbf{val}$ undoes $\mathbf{wrap}$ or $\mathbf{snoc}$, reads the front with $\mathbf{val}$ and combines by $\mathbf{embed}$ or $\mathbf{op}$, so its converse runs that backwards
(b)	$= ([\mathbf{wrap}, \mathbf{snoc}]^\circ \mathbf{Digit}^{+-}(\neg \mathbf{xDigit}(\mathbf{val}))) [\mathbf{embed}, \mathbf{op}]^\circ$	undoing $\mathbf{wrap}$ or $\mathbf{snoc}$ first gives two cases: a decimal and its last digit read by $\mathbf{val}$ then $\mathbf{op}$, or one digit read by $\mathbf{embed}$
(c)	$= (\mathbf{wrap}^\circ \mathbf{embed})^\circ \cup (\mathbf{snoc}^\circ \neg \mathbf{xDigit}(\mathbf{val}) \mathbf{op})^\circ$	the converse of each case is its steps undone in reverse: undo $\mathbf{op}$, undo reading the front, then $\mathbf{snoc}$; or undo $\mathbf{embed}$, then $\mathbf{wrap}$
(d)	$= (\mathbf{embed}^\circ \mathbf{wrap}) \cup (\mathbf{op}^\circ \neg \mathbf{xDigit}(\mathbf{val})^\circ \mathbf{snoc})$	undoing the read of the front while keeping the last digit is $\mathbf{val}^\circ$ on the front alone
(e)	$= (\mathbf{embed}^\circ \mathbf{wrap}) \cup (\mathbf{op}^\circ \neg \mathbf{xDigit}(\mathbf{val}^\circ) \mathbf{snoc})$	

(6.1b)

$$\mathbf{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$$

$\mathbf{op}^\circ$ splits a number into its quotient and remainder by 10

$$(\exists p. 0 < \pi_1 p \wedge \mathbf{op}^\circ(m, p)) \iff 10 \leq m$$

$\mathbf{op}^\circ$ gives a pair with a nonzero first component exactly at the numbers with two or more digits

$$(\exists d. \mathbf{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$$

$\mathbf{embed}^\circ$ gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

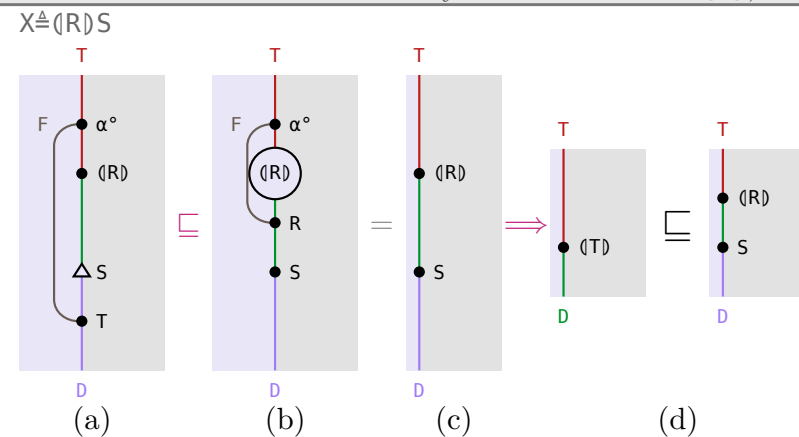
$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation
$\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$ (6.2)	a relation X that contains one unfolding of the fold's recursion at X contains the fold
$X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$ (6.3)	a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

$$F(S) T \sqsubseteq R S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S \quad (6.2c)$$

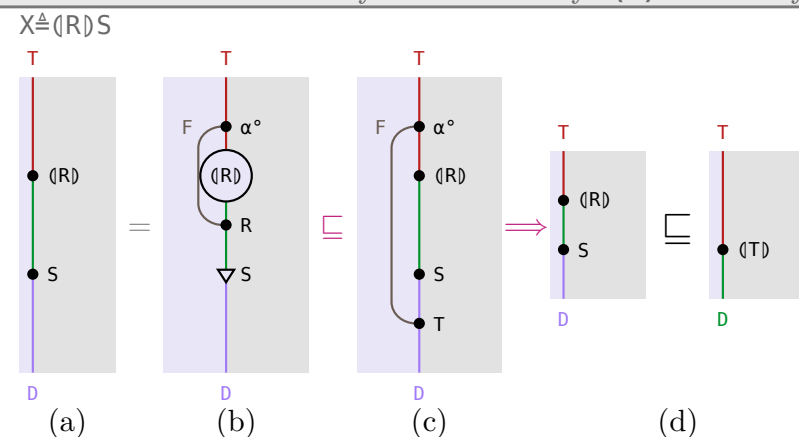
if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S



(a)	$\alpha^\circ F(\langle R \rangle) F(S) T$	$F(S) T \sqsubseteq R S$
(b)	$\sqsubseteq \alpha^\circ F(\langle R \rangle) R S$	$F(\langle R \rangle) R = \alpha(\langle R \rangle)$, $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce
(c)	$= \langle R \rangle S$	$\langle R \rangle S$ is a prefix point of $\alpha^\circ F(-) T$, and $\langle T \rangle$ is the least one — (6.2) $\alpha^\circ F(X) T \sqsubseteq X \Rightarrow \langle T \rangle \sqsubseteq X$
(d)	$\Rightarrow \alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$	

$$R S \sqsubseteq F(S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle \quad (6.2d)$$

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



(a)	$\langle R \rangle S$	$\alpha(\langle R \rangle) = F(\langle R \rangle) R$, $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce
(b)	$= \alpha^\circ F(\langle R \rangle) R S$	$R S \sqsubseteq F(S) T$
(c)	$\sqsubseteq \alpha^\circ F(\langle R \rangle S) T$	$\langle R \rangle S$ is a postfix point of $\alpha^\circ F(-) T$, and $\langle T \rangle$ is the greatest one — (6.3) $X \sqsubseteq \alpha^\circ F(X) T \Rightarrow X \sqsubseteq \langle T \rangle$
(d)	$\Rightarrow \langle R \rangle S \sqsubseteq \alpha^\circ F(\langle R \rangle S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$	

§6.3. Hylomorphisms

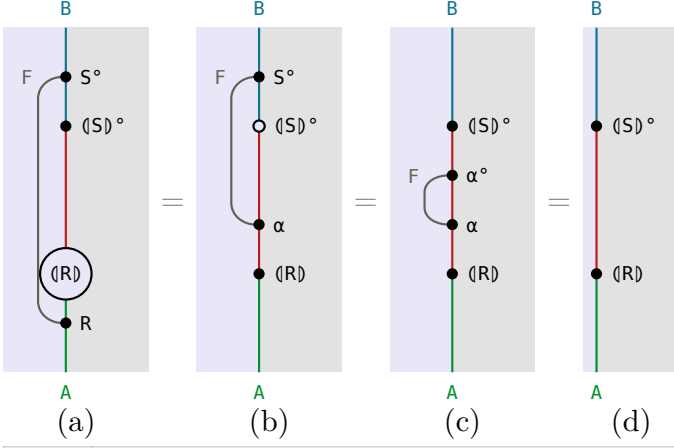
$$(\mathbb{S})^\circ (\mathbb{R}) = (\mu X : S^\circ F(X) R) \quad (6.3a)$$

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$S^\circ F((\mathbb{S})^\circ (\mathbb{R})) R = (\mathbb{S})^\circ (\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((\mathbb{S})^\circ (\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial

the body at $(\mathbb{S})^\circ (\mathbb{R})$

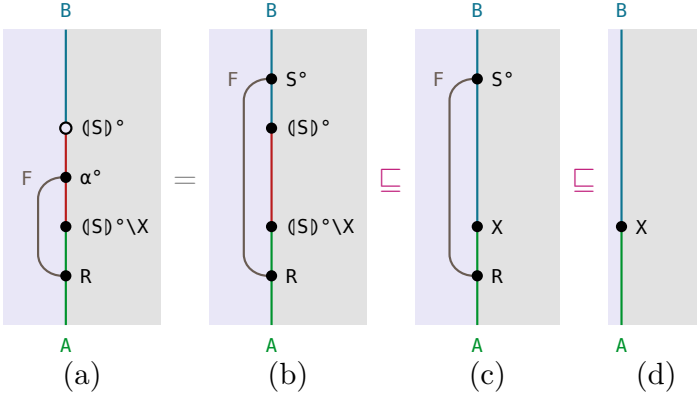


(a)	$S^\circ F((\mathbb{S})^\circ) F((\mathbb{R})) R$	$F((\mathbb{R})) R = \alpha(\mathbb{R}) \triangleq \text{reduce}$
(b)	$= S^\circ F((\mathbb{S})^\circ) \alpha (\mathbb{R})$	$(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ) \triangleq \text{reduce, relator-laws}$
(c)	$= (\mathbb{S})^\circ \alpha^\circ \alpha (\mathbb{R})$	$\alpha \text{ iso}$
(d)	$= (\mathbb{S})^\circ (\mathbb{R})$	

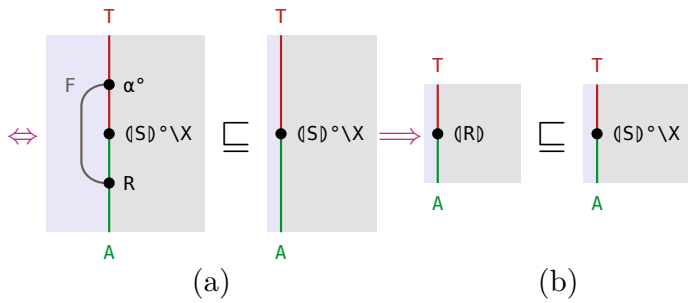
$$S^\circ F(X) R \sqsubseteq X \implies (\mathbb{S})^\circ (\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ (\mathbb{R})$ is included in X if X satisfies the associated recursion inequation

$Y := (\mathbb{S})^\circ \setminus X$



(a)	$(\mathbb{S})^\circ \alpha^\circ F((\mathbb{S})^\circ \setminus X) R$	$(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ)$
(b)	$= S^\circ F((\mathbb{S})^\circ ((\mathbb{S})^\circ \setminus X)) R$	$(\mathbb{S})^\circ ((\mathbb{S})^\circ \setminus X) \sqsubseteq X \text{ — adj-all}$
(c)	$\sqsubseteq S^\circ F(X) R$	$S^\circ F(X) R \sqsubseteq X$
(d)	$\sqsubseteq X$	$S \cdot \neg S \text{ — adj-all}$

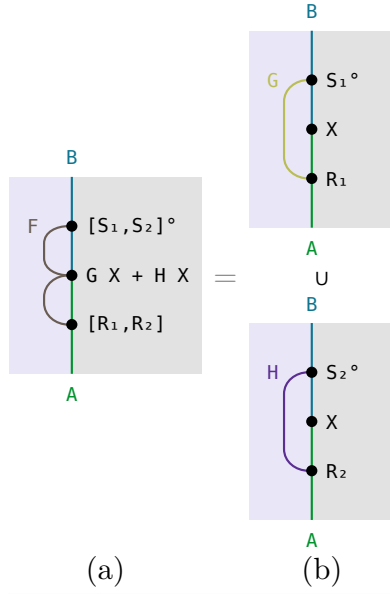


(a)	$\Leftrightarrow S^\circ F(X) R \sqsubseteq X \implies \alpha^\circ F((\mathbb{S})^\circ \setminus X) R \sqsubseteq (\mathbb{S})^\circ \setminus X$	$(\mathbb{R}) = (\mu X : \alpha^\circ F(X) R) \triangleq \text{reduce, (6.2b), adj-all}$
(b)	$\Rightarrow S^\circ F(X) R \sqsubseteq X \implies (\mathbb{R}) \sqsubseteq (\mathbb{S})^\circ \setminus X$	

$$(\forall x. \forall y. \forall X. F(X) = G(X) + H(X)) \Rightarrow ([S_1, S_2]^\circ \llbracket [R_1, R_2] \rrbracket) = (\mu X : (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

the body of Theorem 6.2 at $F(X)=G(X)+H(X)$

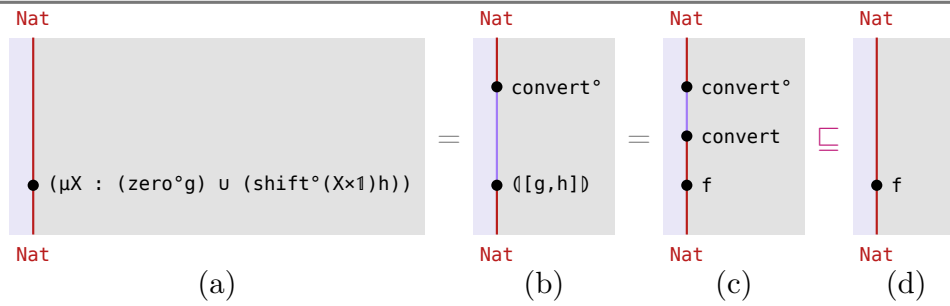


(a)	$[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$	$[S, T] = (S+T)\nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$; the $+$ stands for the codiagonals ∇°, ∇ , which are not drawn
(b)	$= (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)$	$\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct

§6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h)) \sqsubseteq f \quad (6.4a)$$

once f sends **zero** to g and turns **shift** into h , the recursion that halves the argument at each step computes f



(a)	$(\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$	$\text{convert}^\circ \llbracket [g, h] \rrbracket = (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$
(b)	$= \text{convert}^\circ \llbracket [g, h] \rrbracket$	$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow \text{convert } f = \llbracket [g, h] \rrbracket$
(c)	$= \text{convert}^\circ \text{convert } f$	Simple(convert) $1 f = f$
(d)	$\sqsubseteq f$	

$$(\mu X : (\text{zero}^\circ \text{one}) \cup (\text{shift}^\circ (X \times 1) \text{op}(a))) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes a to the power b

$\text{zero } \text{exp}(a) = \text{one}$

a to the power 0 is 1

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

a to the power $2n+d$ is $\text{op}(a)$ of a to the power n and d

$$(\mu X : (\text{zero}^\circ \text{zero}) \cup (\text{shift}^\circ (X \times 1) \text{op}(b))) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves a at each step computes the remainder of a by b

$\text{zero } \text{mod}(b) = \text{zero}$

$0 \text{ mod } b$ is 0

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

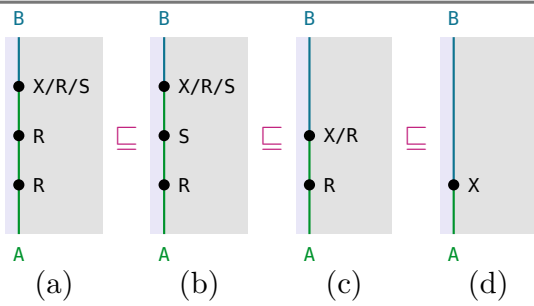
$(2a+d) \text{ mod } b$ is $\text{op}(b)$ of $a \text{ mod } b$ and d

§6.5. Unique fixed points

$$\text{inductive}(S) \wedge R \ R \sqsubseteq S \ R \implies \text{inductive}(R)$$

(6.5a)

R is inductive when $RR \sqsubseteq SR$ for an inductive S



(a)	$(X / R / S) R R$	$RR \sqsubseteq SR$
(b)	$\sqsubseteq (X / R / S) S R$	division by S
(c)	$\sqsubseteq (X / R) R$	division by R
(d)	$\sqsubseteq X$	

$$\text{member} = \emptyset$$

(6.5b)

a constant relator records no elements, so its membership is empty

$$\text{member} = [\text{member}, \text{member}]$$

a member of $F+G$ is a member of whichever summand the value is in

$$\text{member} = (\pi_1 \text{ member}) \cup (\pi_2 \text{ member})$$

a member of $F \times G$ is a member of either component

$$\text{inductive}(S \text{ member}) \wedge X = S F(X) R \wedge Y = S F(Y) R \implies X = Y$$

(6.5c)

when $S \text{ member}$ is inductive, two solutions X, Y of $X = SF(X)R$ are equal

$$\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S F(X) R \sqsubseteq X \implies \text{Entire}(X)$$

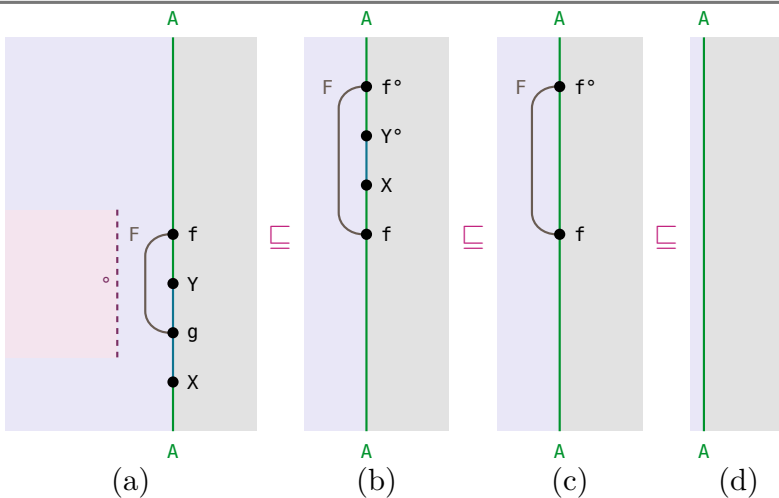
(6.5d)

when $S \text{ member}$ is inductive and S, R are entire, a pre-fixed point of $X \mapsto SF(X)R$ is entire

$$\text{inductive}(g \text{ member}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = g F(X) f \implies \text{Map}(X)$$

(6.5e)

when $g \text{ member}$ is inductive and f, g are maps, the solution of $X = gF(X)f$ is a map

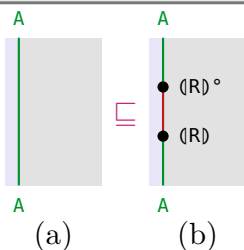


(a)	$(g F(Y) f)^\circ X$	$X = gF(X)f, g \text{ simple}$
(b)	$\sqsubseteq f^\circ F(Y^\circ X) f$	$Y^\circ X \sqsubseteq 1$
(c)	$\sqsubseteq f^\circ f$	$f \text{ simple}$
(d)	$\sqsubseteq 1$	

$$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R \implies 1 \sqsubseteq (R)^\circ (R)$$

(6.5f)

when $R^\circ \text{ member}$ is inductive and R is surjective, the fold (R) is surjective



(a)	1	(6.5d) at $S = R^\circ$
(b)	$\sqsubseteq (R)^\circ (R)$	

Map(f) ∧ R f ⊆ F(f) α ∧ 1 ⊆ R° R ⇒ f° = ⟨R⟩ (6.5g)		
a map f with Rf⊆F(f)α, for a surjective R, has the fold ⟨R⟩ as its converse		
T • ⟨R⟩ A (a) T • f° A (b) ⊆		
(a)	⟨R⟩	shunting f, fusion, Rf⊆F(f)α
(b)	⊆ f°	

T • f° A (a) T • ⟨R⟩ A (b) ⊆		
(a)	f°	(6.5f) and ⟨R⟩f⊆1
(b)	⊆ ⟨R⟩	

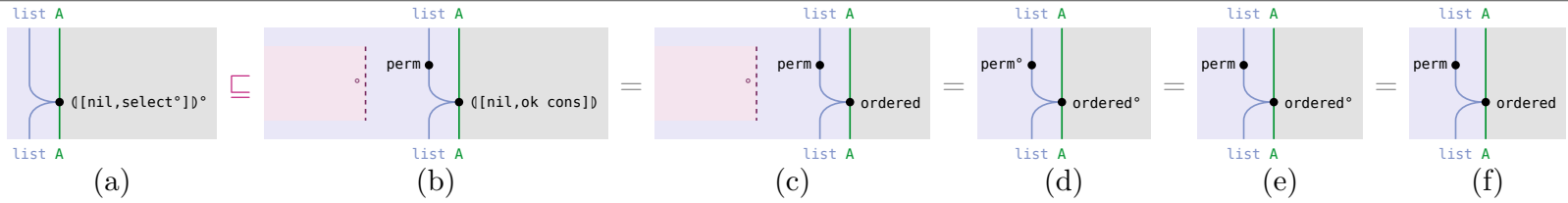
$\text{Map}(f) \wedge R \ f \sqsubseteq F(f) \ \alpha \implies R^\circ \text{ member} \sqsubseteq f \ \alpha^\circ \text{ member } f^\circ$ (6.5h)		
$R^\circ \text{ member}$ is below $\alpha^\circ \text{ member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{ member}$ is		
(a) (b) (c)		
(a)	$R^\circ \text{ member}$	$\text{Map}(f) \wedge R \ f \sqsubseteq F(f) \ \alpha \implies R^\circ \sqsubseteq f \ \alpha^\circ F(f^\circ)$ $F(R) \text{ member} \sqsubseteq \text{member } R$
(b)	$\sqsubseteq f \ \alpha^\circ F(f^\circ) \text{ member}$	
(c)	$\sqsubseteq f \ \alpha^\circ \text{ member } f^\circ$	

§6.6. Sorting by selection

$\text{ordered} = \langle [\text{nil}, \text{ok cons}] \rangle$ (6.6a)		
a list is sorted exactly when the fold that rebuilds it passes ok at every cons , i.e. each head is R -below every element after it		
(a) (b)		
(a)	ordered	fold uniqueness
(b)	$= \langle [\text{nil}, \text{ok cons}] \rangle$	

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq \text{perm ordered} \quad (6.6b)$$

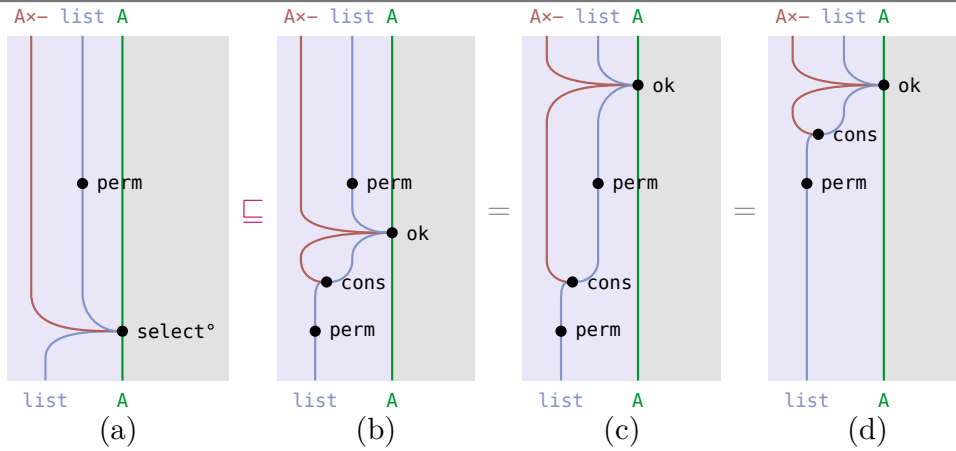
every output of unfolding the input by **select** is a sorted permutation of it



(a)	$[\text{nil}, \text{select}^\circ]^\circ$	$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq$
(b)	$\sqsubseteq ([\text{nil}, \text{ok cons}]^\circ \text{ perm})^\circ$	$([\text{nil}, \text{ok cons}]^\circ \text{ perm})^\circ$
(c)	$= (\text{ordered perm})^\circ$	$\text{ordered} = [\text{nil}, \text{ok cons}]^\circ$
(d)	$= \text{perm}^\circ \text{ ordered}^\circ$	$(R S)^\circ = S^\circ R^\circ$
(e)	$= \text{perm} \text{ ordered}^\circ$	$\text{perm}^\circ = \text{perm}$
(f)	$= \text{perm ordered}$	$\text{ordered}^\circ = \text{ordered}$

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{ select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

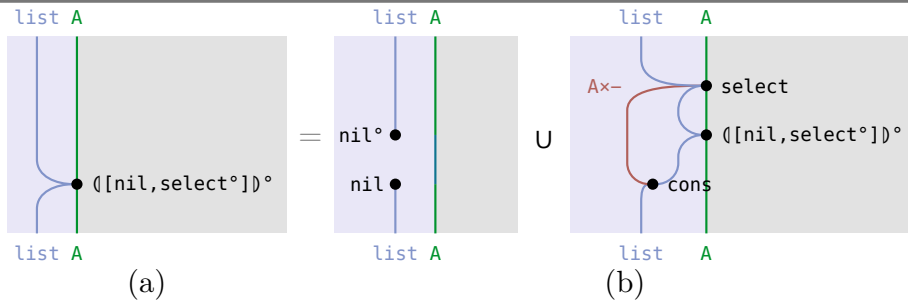
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



(a)	$(1 \times \text{perm}) \text{ select}^\circ$	$\text{select}^\circ \sqsubseteq \text{ok cons perm}$
(b)	$\sqsubseteq (1 \times \text{perm}) \text{ ok cons perm}$	$\text{ok} (1 \times \text{perm}) = (1 \times \text{perm}) \text{ ok}$
(c)	$= \text{ok} (1 \times \text{perm}) \text{ cons perm}$	$\text{cons perm} = (1 \times \text{perm}) \text{ cons perm}$
(d)	$= \text{ok cons perm}$	

$$([\text{nil}, \text{select}^\circ])^\circ = (\text{nil}^\circ \text{ nil}) \cup (\text{select} (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{ cons}) \quad (6.6d)$$

selection sort returns **[]** on **[]**, and otherwise selects **(a,y)**, sorts **y** and conses **a** back on

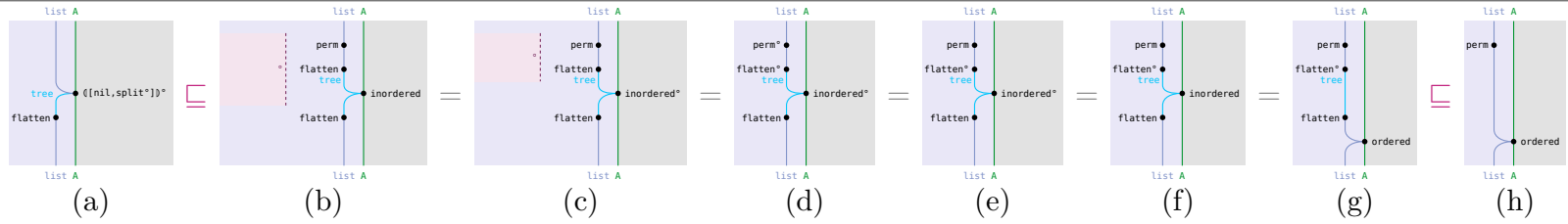


(a)	$[\text{nil}, \text{select}^\circ]^\circ$	unfolding the converse of a fold
(b)	$= (\text{nil}^\circ \text{ nil}) \cup (\text{select} (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{ cons})$	

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ([\text{nil}, \text{split}^\circ])^\circ \quad (6.6e)$$

flatten $\sqsubseteq$ perm ordered

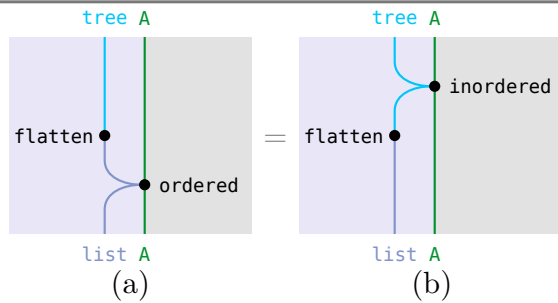
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



(a)	$[\text{nil}, \text{split}^\circ]^\circ \text{ flatten}$	$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ([\text{nil}, \text{split}^\circ])^\circ$
(b)	$\sqsubseteq (\text{inordered flatten perm})^\circ \text{ flatten}$	$\text{flatten} \sqsubseteq (\text{inordered flatten perm})^\circ \text{ flatten}$
(c)	$= (\text{flatten perm})^\circ \text{ inordered}^\circ \text{ flatten}$	$(R S)^\circ = S^\circ R^\circ$
(d)	$= \text{perm}^\circ \text{ flatten}^\circ \text{ inordered}^\circ \text{ flatten}$	$(R S)^\circ = S^\circ R^\circ$
(e)	$= \text{perm} \text{ flatten}^\circ \text{ inordered}^\circ \text{ flatten}$	$\text{perm}^\circ = \text{perm}$
(f)	$= \text{perm} \text{ flatten}^\circ \text{ inordered} \text{ flatten}$	$\text{inordered}^\circ = \text{inordered}$
(g)	$= \text{perm} \text{ flatten}^\circ \text{ flatten} \text{ ordered}$	$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow$ $\text{flatten ordered} = \text{inordered flatten}$
(h)	$\sqsubseteq \text{perm ordered}$	$\text{flatten}^\circ \text{ flatten} \sqsubseteq 1$ $1 f = f$

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten} \quad (6.6f)$$

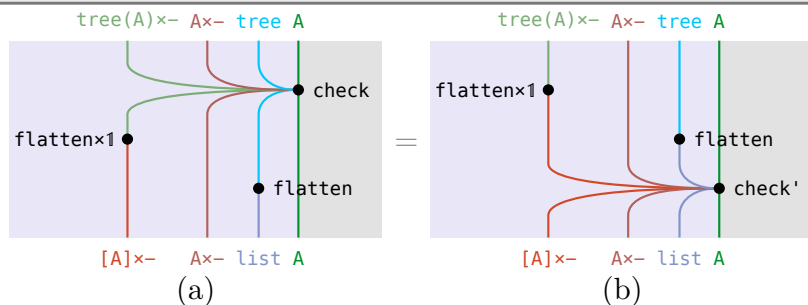
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



(a)	flatten ordered	R transitive: a flattened tree is ordered iff every node passes check
(b)	$= \text{inordered flatten}$	

$$\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \quad (6.6g)$$

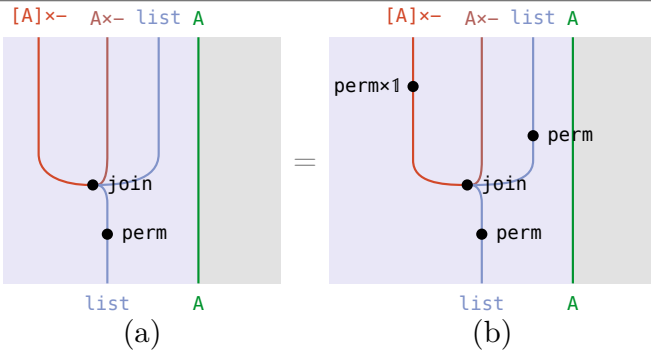
testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



(a)	$\text{check} (\text{flatten} \times (1 \times \text{flatten}))$	$b \text{ intree } x \iff b \text{ inlist } \text{flatten}(x)$
(b)	$= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}'$	

$$\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm} \quad (6.6h)$$

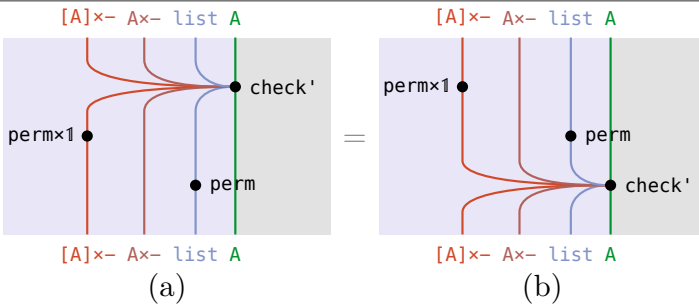
permuting the list $x \mathrel{++} [a] \mathrel{++} y$ is the same as first permuting x and y and then permuting the joined list



(a)	join perm	perm is transitive and closed under ++
(b)	= (perm × (1 × perm)) join perm	

$$\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{ check}' \quad (6.6i)$$

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

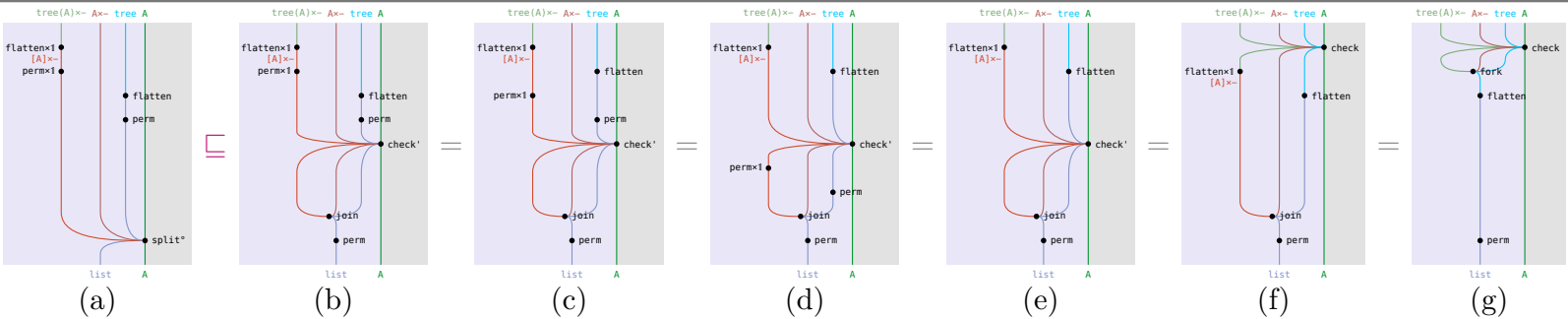


(a)	check' (perm × (1 × perm))	$b \text{ inlist } x \iff b \text{ inlist perm}(x)$
(b)	= (perm × (1 × perm)) check'	

$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \implies ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ \sqsubseteq \text{check fork} \quad (6.6j)$$

flatten perm

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



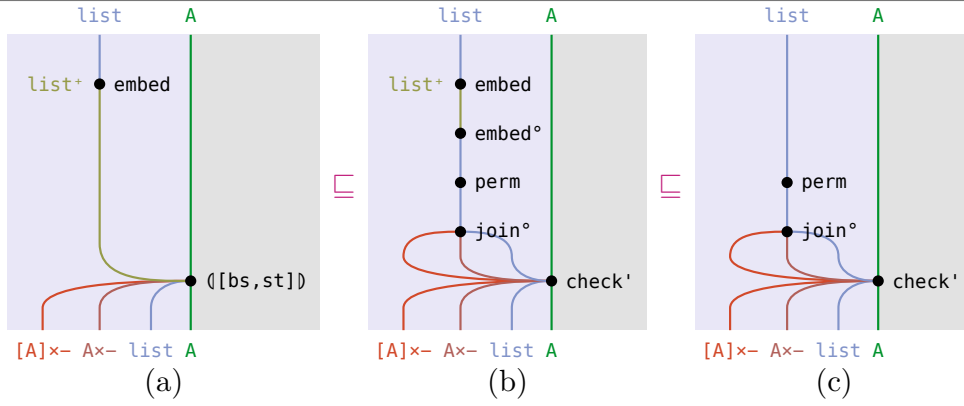
(a)	$((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ$	$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$
(b)	$\sqsubseteq ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ check}' \text{ join perm}$	$(f \times (1 \times f)) (g \times (1 \times g)) = (f g) \times (1 \times (f g))$
(c)	$= (\text{flatten} \times (1 \times \text{flatten})) (\text{perm} \times (1 \times \text{perm})) \text{ check}' \text{ join perm}$	$\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{ check}'$
(d)	$= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' (\text{perm} \times (1 \times \text{perm})) \text{ join perm}$	$\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm}$
(e)	$= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \text{ join perm}$	$\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{ check}'$
(f)	$= \text{check} (\text{flatten} \times (1 \times \text{flatten})) \text{ join perm}$	$\text{fork flatten} = (\text{flatten} \times (1 \times \text{flatten})) \text{ join}$
(g)	$= \text{check fork flatten perm}$	

$bs \sqsubseteq \text{wrap perm join}^\circ \text{check}' \wedge (1 \times (\text{perm join}^\circ \text{check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{check}' \Rightarrow \text{embed}$ (6.6k)

$([bs, st]) \sqsubseteq \text{perm join}^\circ \text{check}'$

turning a list into a non-empty list and folding it with any $[bs, st]$ meeting the two conditions below gives only

checked triples (x, a, y) whose $\text{join } x \# [a] \# y$ is a permutation of the list

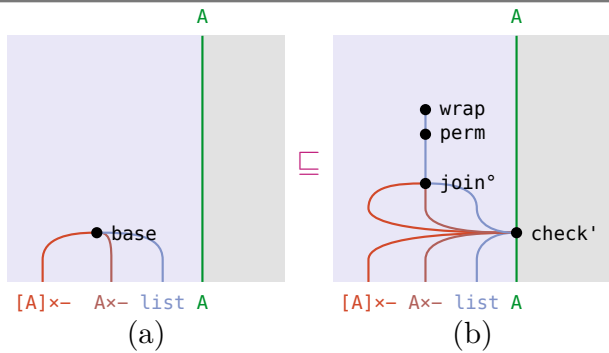


(a)	$\text{embed } ([bs, st])$	$bs \sqsubseteq \text{wrap perm join}^\circ \text{check}' \wedge (1 \times (\text{perm join}^\circ \text{check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{check}' \Rightarrow \text{embed } ([bs, st]) \sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$
(b)	$\sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$	$\text{embed embed}^\circ \text{perm join}^\circ \text{check}' \sqsubseteq \text{perm join}^\circ \text{check}'$
(c)	$\sqsubseteq \text{perm join}^\circ \text{check}'$	

$\text{base} \sqsubseteq \text{wrap perm join}^\circ \text{check}'$

(6.6l)

base(a) is a **checked triple** whose **join** is a permutation of the one-element list $[a]$

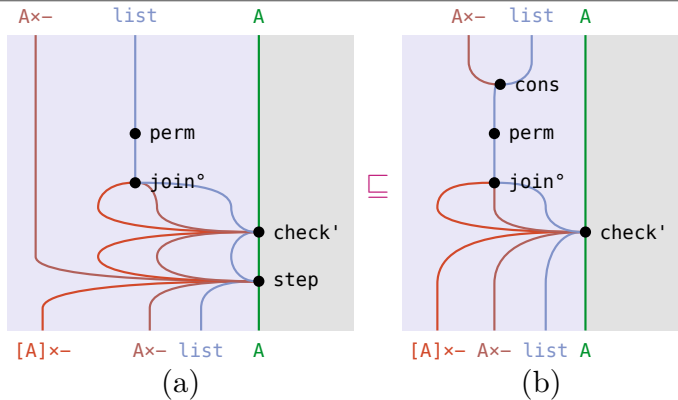


(a)	base	$[] \# [a] \# [] = [a]$, and check' holds on empty lists
(b)	$\sqsubseteq \text{wrap perm join}^\circ \text{check}'$	

$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \Rightarrow (1 \times (\text{perm join}^\circ \text{check}')) \text{ step} \sqsubseteq \text{cons perm join}^\circ \text{check}'$ (6.6m)

if (x, b, y) **is a checked triple** for a permutation of l , **then** $\text{step}(a, (x, b, y))$ **is a checked triple** for a

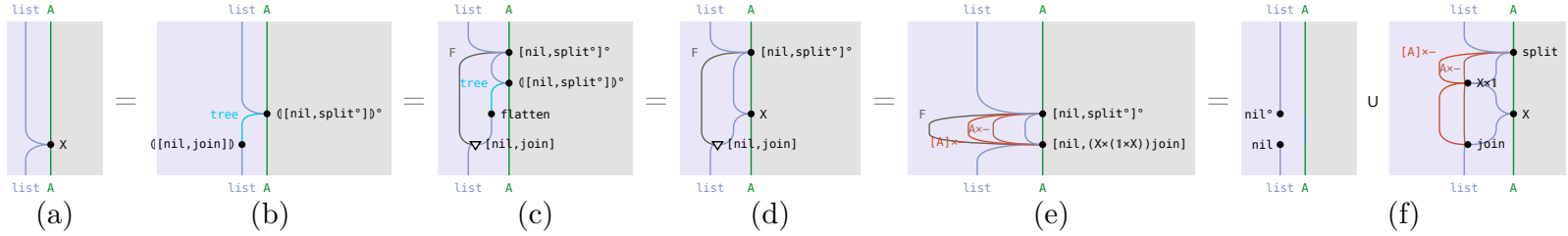
permutation of $[a] \# l$



(a)	$(1 \times (\text{perm join}^\circ \text{check}')) \text{ step}$	aRb puts a in front of x ; otherwise bRa and a goes in front of y
(b)	$\sqsubseteq \text{cons perm join}^\circ \text{check}'$	

$$X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten} \Rightarrow X = (\text{nil}^\circ \text{ nil}) \cup (\text{split} (X \times (1 \times X)) \text{ join}) \quad (6.6n)$$

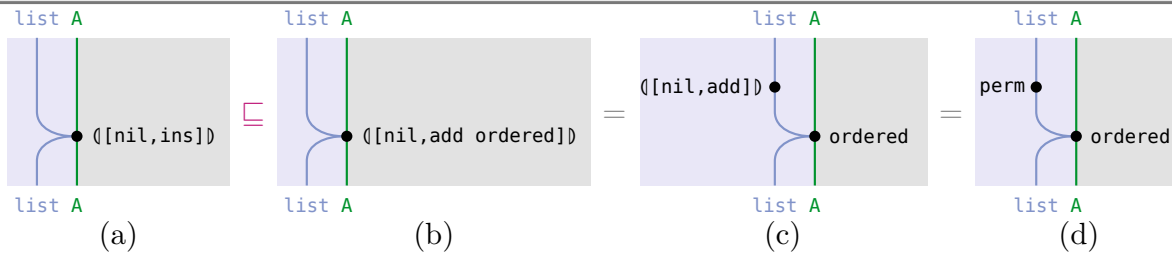
quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a



(a)	X	$X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$
(b)	$= ([\text{nil}, \text{split}^\circ])^\circ ([\text{nil}, \text{join}])$	$S^\circ F((S)^\circ (R)) R = (S)^\circ (R)$
(c)	$= [\text{nil}, \text{split}^\circ]^\circ F([\text{nil}, \text{split}^\circ])^\circ \text{ flatten} [\text{nil}, \text{join}]$	$X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$
(d)	$= [\text{nil}, \text{split}^\circ]^\circ F(X) [\text{nil}, \text{join}]$	$F(S) [T, U] = [T, (S \times (1 \times S)) U]$
(e)	$= [\text{nil}, \text{split}^\circ]^\circ [\text{nil}, (X \times (1 \times X)) \text{ join}]$	$[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$
(f)	$= (\text{nil}^\circ \text{ nil}) \cup (\text{split} (X \times (1 \times X)) \text{ join})$	

$$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered} \Rightarrow ([\text{nil}, \text{ins}]) \sqsubseteq \text{perm ordered} \quad (6.6o)$$

for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations

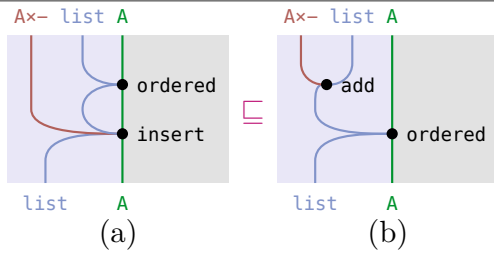


(a)	$[\text{nil}, \text{ins}]$	$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered} \Rightarrow$
(b)	$\sqsubseteq [\text{nil}, \text{add ordered}]$	$[\text{nil}, \text{ins}] \sqsubseteq [\text{nil}, \text{add ordered}]$
(c)	$= [\text{nil}, \text{add}] \text{ ordered}$	$[\text{nil}, \text{add}] \text{ ordered} = [\text{nil}, \text{add ordered}]$
(d)	$= \text{perm ordered}$	$\text{perm} = [\text{nil}, \text{add}]$

$$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \wedge (\forall a. \forall b. \forall c. \quad (6.6p)$$

$$R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \Rightarrow (1 \times \text{ordered}) \text{ insert} \sqsubseteq \text{add ordered}$$

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



(a)	$(1 \times \text{ordered}) \text{ insert}$	insert splices a in, and R transitive and total
(b)	$\sqsubseteq \text{add ordered}$	

§6.7. Closure

$$R^* = (\mu X : 1 \cup (X R)) \quad (6.7a)$$

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R***

$$1 \sqsubseteq X \wedge X X \sqsubseteq X \Rightarrow R \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X \quad (6.7b)$$

for a preorder **X**, **X** contains **R** exactly when it contains **R***

$$1 \sqsubseteq R^* \quad (6.7c)$$

R* is reflexive

(a)	1	
(b)	$\sqsubseteq 1 \cup (R R^*)$	$R \sqsubseteq R \cup S$
(c)	$= R^*$	$1 \cup (R R^*) = R^*$

$R \subseteq R^*$ (6.7d)		
R^* contains R		
(a)	R	
(b)	$= R$	$f \ 1 = f$
(c)	$\sqsubseteq R \ R^*$	$1 \subseteq R^*$
(d)	$\sqsubseteq R^*$	$R \ R^* \subseteq R^*$

$R^* \ R^* \subseteq R^*$ (6.7e)		
R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup R X$		
(a)	$(1 \cup (R \ (R^* / R^*))) \ R^*$	
(b)	$= R^* \cup (R \ (R^* / R^*) \ R^*)$	$(P \cup Q) T = P T \cup Q T$ — composition distributes over $\cup$
(c)	$\sqsubseteq R^* \cup (R \ R^*)$	$(S/S) S \subseteq S$
(d)	$\sqsubseteq R^*$	$R R^* \subseteq R^*$ — (6.7d)

$1 \subseteq X \wedge X X \subseteq X \wedge R \subseteq X \implies R^* \subseteq X$ (6.7f)		
a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup R X$, so it contains R^*		
(a)	$1 \cup (R \ X)$	
(b)	$\sqsubseteq X \cup (X \ X)$	$1 \subseteq X, R \subseteq X$
(c)	$\sqsubseteq X$	$X X \subseteq X$

$\frac{R^*}{\exists} = \left(\frac{1}{\exists}, \frac{R \ R^*}{\exists} \right) \text{ cup}$ (6.7g)		
the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors		
(a)	$\frac{R^*}{\exists}$	$1 \cup (R \ R^*) = R^*$
(b)	$= 1 \cup (R \ R^*)$	$\frac{R \cup S}{\exists} = \left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{ cup}$
(c)	$= \left(\frac{1}{\exists}, \frac{R \ R^*}{\exists} \right) \text{ cup}$	

$R - \emptyset = R$ (6.7h)	
taking nothing away leaves R	
$R \cup (S - R) = R \cup S$	
beside R , only the part of S outside R adds anything	
$R - (S \cup T) = R - S - T$	
taking away S and then T takes away $S \cup T$	
$(R \cup S) - T = (R - T) \cup (S - T)$	
taking T away from a union takes it away from each part	

$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$ (6.7i)	
the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ	

$S \ R^* = (\mu X : S \cup (X \ R))$ (6.7j)	
S followed by any number of R steps is the least X containing S and closed under a further R step	
$R^* \ S = (\mu X : S \cup (R \ X))$	
any number of R steps followed by S is the least X containing S and closed under an R step in front	

$\theta(\emptyset, S) = S \ R^*$ (6.7k)		
started with nothing found and S to explore, θ computes $S R^*$		
(a)	$\theta(\emptyset, S)$	
(b)	$\triangleq \emptyset \cup (\mu X : S \cup ((X \ R) - \emptyset))$	
(c)	$= \emptyset \cup (\mu X : S \cup (X \ R))$	$R - \emptyset = R$
(d)	$= (\mu X : S \cup (X \ R))$	$\emptyset \cup R = R$
(e)	$= S \ R^*$	$S \ R^* = (\mu X : S \cup (X \ R))$

$$\theta(P, \emptyset) = P$$

(6.71)

with nothing left to explore, θ returns what it has found

(a)	$\theta(P, \emptyset)$	
(b)	$\triangleq P \cup (\mu X : \emptyset \cup ((X \ R) - P))$	
(c)	$= P \cup \emptyset$	$(\mu X : \emptyset \cup ((X \ R) - P)) = \emptyset$
(d)	$= P$	$R \cup \emptyset = R$